

StreamSense: A Content- and Resource-Aware Edge Agent for Adaptive Bitrate Streaming

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Abstract

Adaptive Bitrate (ABR) decision-making is essential for optimizing Quality of Experience (QoE) in video streaming. Traditional ABR methods typically lack fine-grained content understanding and face challenges in balancing multi-dimensional QoE objectives under real-world edge device constraints. To overcome these limitations, we propose **StreamSense Agent**, a novel framework that reformulates ABR as a sequential decision-making process informed by content and network awareness. Our approach introduces three key innovations: (1) a Dynamic Content-Aware Entity Graph (D-CAEG) for structured video representation, (2) a composite QoE-driven optimization objective integrating perceptual quality, rebuffering, smoothness, and computational cost, and (3) a hierarchical resource-aware architecture enabling real-time control through an efficient lightweight actor. We further adopt an offline-to-online co-adaptation training paradigm with federated learning for continuous adaptation. Since effective ABR control requires the same core competencies as general sequential decision-making—multi-step planning across future video chunks, dynamic information gathering from heterogeneous network signals, and adaptive strategy adjustment under resource constraints—we validate these transferable capabilities on five established benchmarks. Specifically, multi-hop search tasks (BrowseComp, BrowseComp-ZH) mirror the integration of dispersed network-trace and content-complexity signals into a bitrate plan; complex navigation tasks (WebWalkerQA) correspond to sequential buffer-state traversal under volatile bandwidth; and information-extraction tasks (Xbench-DS, SimpleQA) reflect the precision required when reading real-time streaming telemetry. StreamSense Agent achieves state-of-the-art results, attaining accuracy of 16.7% on BrowseComp, 30.5% on BrowseComp-ZH, 42.1% on

Xbench-DS, and 54.2% on WebWalkerQA. Deployment-oriented evaluations further confirm its computational efficiency, achieving an inference throughput of 12.5 requests per second with a memory footprint of only 2,100 MB under edge-like constraints, yielding a normalized QoE score of 0.92. Comprehensive ablation studies validate the necessity of each proposed component. This work establishes a new paradigm for intelligent, edge-deployable bitrate adaptation.

1 Introduction

The widespread adoption of high-quality video streaming has fundamentally transformed digital content consumption, making adaptive bitrate (ABR) algorithms central to user experience (Bentaleb et al., 2019; Seufert et al., 2015). These algorithms must dynamically adjust video quality in response to fluctuating network conditions to optimize Quality of Experience (QoE). However, modern streaming demands—including diverse content types, stringent edge-device resource constraints, and multi-dimensional QoE objectives—expose the limitations of conventional ABR methods. Traditional approaches often treat video as a uniform byte sequence, lacking fine-grained content understanding (Dobrian et al., 2011), or rely on heuristic rules (Huang et al., 2014; Spiteri et al., 2020; Yin et al., 2015) that are inadequate for robust, long-horizon optimization under real-world deployment constraints.

To address these challenges, we propose **StreamSense Agent**, a novel framework that reformulates ABR decision-making as a content- and network-aware sequential decision-making process. Figure 1 illustrates the key limitations of traditional ABR methods and highlights the integrated, content- and resource-aware design of our proposed approach. Our approach is built upon three core innovations: (1) a **Dynamic Content-Aware Entity Graph (D-CAEG)** that provides a structured and

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queryable representation of video semantics and encoding complexity; (2) a **Composite QoE-Driven Optimization Objective** that integrates perceptual quality, rebuffering avoidance, quality smoothness, and computational cost into a unified reward signal; and (3) a **Hierarchical Resource-Aware Decision Architecture** that enables real-time, low-latency control suitable for edge deployment.

Since effective ABR control fundamentally relies on sequential reasoning and long-horizon planning—the agent must anticipate future content demands, navigate trade-offs among competing objectives, and adapt to dynamic environments—we validate these core capabilities on five established benchmarks that probe multi-step planning, complex navigation, and information retrieval. Concretely, the multi-hop search capability evaluated by BrowseComp and BrowseComp-ZH directly corresponds to the ABR agent’s need to integrate dispersed network-trace and content-complexity signals into a coherent bitrate plan; the sequential navigation tasks in WebWalkerQA mirror the buffer-state traversal that ABR agents must perform under volatile bandwidth; and the information-extraction tasks in Xbench-DS and SimpleQA reflect the precision required when reading multi-dimensional streaming telemetry to inform each successive bitrate decision.

StreamSense Agent achieves state-of-the-art performance, attaining an accuracy of 16.7% on BrowseComp and 30.5% on BrowseComp-ZH, surpassing both open-source and larger-parameter models. On WebWalkerQA, it achieves 54.2%, demonstrating strong multi-step navigation and planning capabilities. Crucially, deployment-oriented evaluations under simulated edge constraints confirm an inference throughput of 12.5 requests per second and a memory footprint of only 2,100 MB, validating its practical suitability for resource-constrained streaming environments. The rest of this paper is organized as follows: Section 2 reviews related work, Section 3 details our methodology, Section 4 presents experimental results and ablation studies, Section C provides a discussion of training dynamics and deployment scenarios, and Section 5 concludes.

2 Related Work

StreamSense Agent draws upon advances in reasoning models, retrieval-augmented generation, autonomous agents (Xu et al., 2025), and adaptive

bitrate streaming. In this section, we review these four pillars and position our contributions within this landscape.

2.1 Autonomous Agents and Benchmarks

Recent research has extended agent capabilities to complex web-browsing and sequential decision-making tasks (Nakano et al., 2021; Yao et al., 2023; Bao et al., 2026; Song et al., 2025). Models such as DeepDive, WebSailor, and ASearcher exemplify this direction. Their development is often supported by sophisticated data-synthesis methods like InfoSeek, which generates long-horizon browsing trajectories via knowledge-graph question synthesis and scalable reinforcement learning. Further frameworks integrate *think-search-write* loops in real-web environments, including Search-o1, Search-R1, DeepResearcher, and the multi-agent WebThinker. Other approaches, such as AgentFly, achieve advanced performance through memory-based continual learning without fine-tuning the base model. At the foundation model level, systems such as GLM-4.5 and Kimi-Researcher employ multi-stage post-training with reinforcement learning to strengthen tool use and reasoning. Concurrently, new benchmarks have emerged to evaluate these autonomous agents, including BrowseComp (Wei et al., 2025), HLE, BrowseComp-ZH (Zhou et al., 2025), xBench, and GAIA. These benchmarks present complex tasks requiring the integration of diverse information sources, posing a greater challenge than traditional RAG or knowledge-based QA evaluations. We adopt these established benchmarks to validate the sequential planning and reasoning backbone of StreamSense Agent through an explicit skill-to-ABR mapping: (i) multi-hop search, as evaluated in BrowseComp (Wei et al., 2025) and BrowseComp-ZH (Zhou et al., 2025), corresponds to integrating heterogeneous network measurements and content-complexity signals into a coherent long-horizon bitrate plan; (ii) complex sequential navigation, as tested in WebWalkerQA (Wu et al., 2025), mirrors the agent’s sequential buffer-state traversal under volatile network conditions; and (iii) precise factual retrieval, assessed in Xbench-DS and SimpleQA, reflects the accuracy required when extracting actionable quality metrics from multi-dimensional streaming telemetry. Since these capabilities are foundational to any high-performance ABR agent, benchmark-level excellence provides a rigorous, reproducible proxy for the agent’s core

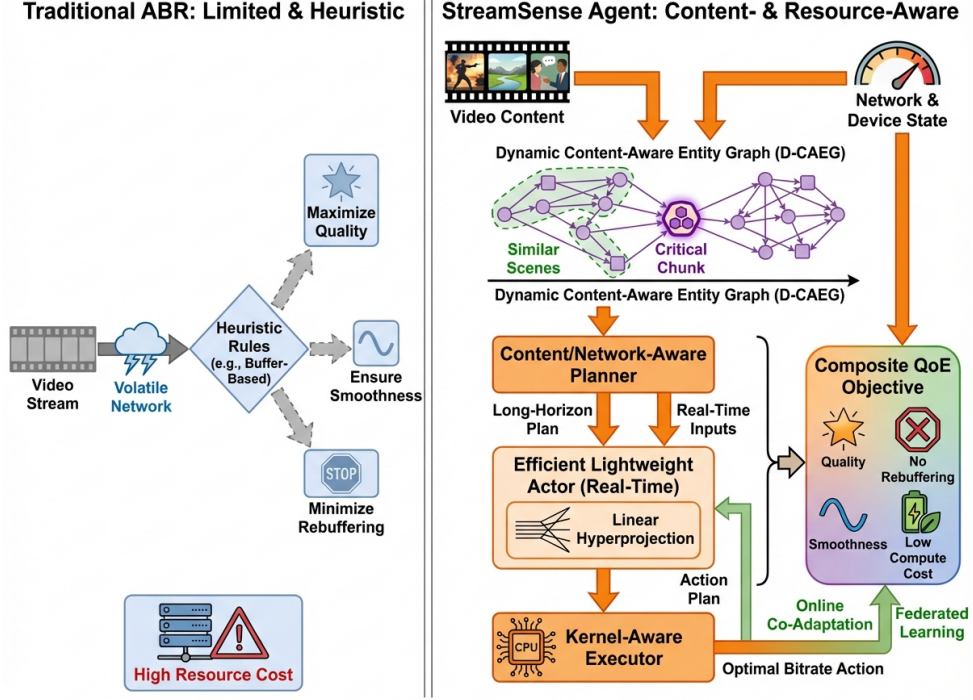


Figure 1: Motivation for StreamSense Agent. Left: Traditional ABR methods rely on heuristic rules with isolated optimization objectives and lack content understanding, leading to suboptimal quality-of-experience under volatile network conditions. Right: StreamSense Agent integrates a Dynamic Content-Aware Entity Graph (D-CAEG) with a hierarchical resource-aware architecture and a unified composite QoE objective, enabling content-aware, long-horizon planning with online federated co-adaptation.

decision-making competence prior to streaming-environment deployment.

2.2 Adaptive Bitrate Streaming

Adaptive bitrate (ABR) algorithms have evolved substantially over the past decade (Stockhammer, 2011; Bentaleb et al., 2019). Classical rule-based approaches, such as BBA (Buffer-Based Approach) (Huang et al., 2014) and BOLA (Spiteri et al., 2020), select bitrate levels based on buffer occupancy thresholds, offering simplicity and low computational overhead but lacking adaptability to complex network dynamics. Throughput-based methods estimate future bandwidth from recent measurements and select the highest sustainable bitrate (Huang et al., 2012); however, they are susceptible to estimation errors under volatile conditions. Model Predictive Control (MPC) approaches formulate ABR as a finite-horizon optimization problem over predicted throughput trajectories (Yin et al., 2015), achieving improved QoE at the cost of higher computational demand. More recently, deep reinforcement learning (DRL) methods—notably Pensieve (Mao et al., 2017) and its successors—have demonstrated that learned policies can outper-

form hand-crafted heuristics (Mnih et al., 2015) by optimizing a multi-term QoE objective end-to-end. Subsequent work has explored multi-agent RL for multi-client scenarios (Lowe et al., 2017), meta-learning for rapid adaptation to unseen network conditions (Finn et al., 2017), and attention-based architectures (Vaswani et al., 2017; Song et al., 2024) for improved temporal modeling. Despite these advances, most existing ABR methods treat video content as a uniform byte stream and do not incorporate fine-grained content semantics into the decision process. Furthermore, few approaches explicitly address the joint optimization of QoE and computational cost under edge device constraints. StreamSense Agent distinguishes itself by integrating structured content understanding via the D-CAEG, a composite QoE objective that includes computational cost, and a hierarchical architecture specifically designed for resource-constrained edge deployment.

3 Methodology

We introduce **StreamSense Agent**, a novel framework for intelligent bitrate adaptation in streaming media. The overall architecture of the Stream-

Sense Agent framework is illustrated in Figure 2. StreamSense Agent formulates the Adaptive Bitrate (ABR) decision problem as a **content- and network-aware sequential decision-making process** under edge resource constraints.

Our approach integrates computer vision for content understanding, reinforcement learning for long-term Quality of Experience (QoE) optimization, and lightweight model design for edge deployment (Bai et al., 2026). The core innovations are: (1) a **Dynamic Content-Aware Entity Graph (D-CAEG)** for structured video representation, (2) a **Composite QoE-Driven Optimization Objective** unifying multi-dimensional streaming objectives, and (3) a **Hierarchical Resource-Aware Decision Architecture** for real-time, low-latency control.

3.1 Dynamic Content-Aware Entity Graph (D-CAEG) Construction

Traditional ABR methods typically treat video content as a uniform or simply segmented byte sequence, lacking fine-grained semantic and complexity modeling. Inspired by knowledge graphs in NLP (Bordes et al., 2013), we construct a **Dynamic Content-Aware Entity Graph (D-CAEG)** $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, providing a structured, queryable representation to guide adaptive decisions.

3.1.1 Node Initialization & Content Feature Extraction.

Given a video asset, we segment it into chunks $\{v_i\}_{i=1}^N$, each corresponding to a short playback duration (e.g., 2–4 seconds). Unlike prior work that treats chunks as opaque byte sequences, we define each node v_i with a rich feature vector:

$$v_i = (\mathbf{f}_i^{vis}, \mathbf{f}_i^{enc}, \mathbf{f}_i^{sem}, meta_i). \quad (1)$$

Here, $\mathbf{f}_i^{vis} \in \mathbb{R}^{d_v}$ represents visual complexity features extracted by a lightweight CNN (e.g., MobileNet (Howard et al., 2017) or ResNet (He et al., 2016)), capturing spatial and temporal entropy. $\mathbf{f}_i^{enc} \in \mathbb{R}^{d_e}$ denotes encoding-related features, including the bitrate ladder $\{b_{i,1}, \dots, b_{i,K}\}$ and predicted quality metrics (Zhong et al., 2025) (e.g., VMAF). $\mathbf{f}_i^{sem} \in \mathbb{R}^{d_s}$ is a semantic embedding (e.g., from a video-language model (Radford et al., 2021; Liang et al., 2026; Zhang et al., 2024)) describing the chunk content. $meta_i$ includes basic metadata such as chunk duration and index.

3.1.2 Edge Formation & Graph Dynamics.

Edges $\mathcal{E}$ model relationships between chunks. We define two primary edge types:

- **Temporal Edges ($\mathcal{E}_T$):** Connect consecutive chunks (v_i, v_{i+1}) , forming the timeline backbone.
- **Semantic-Similarity Edges ($\mathcal{E}_S$):** Connect non-sequential chunks (v_i, v_j) with cosine similarity between $\mathbf{f}_i^{sem}$ and $\mathbf{f}_j^{sem}$ above a threshold τ , enabling anticipation of content transitions and pre-fetching for similar scenes.

The graph is *dynamic* as the agent’s view expands during playback, and edge weights (e.g., similarity) can be updated with real-time user interaction in online learning.

3.1.3 Content-Complexity-Aware Constraint Set.

For each node v_i , we derive a **Content-Complexity-Aware Constraint Set** $\mathcal{C}(v_i)$, encapsulating delivery challenges under resource constraints. It is formulated as the output of an optimization problem seeking the minimal feature set for robust ABR decisions:

$$\begin{aligned} \mathcal{C}(v_i) = \arg \min_{\mathcal{S} \subseteq \{\mathbf{f}_i^{vis}, \mathbf{f}_i^{enc}\}} \|\mathcal{S}\|_0, \\ \text{s.t. } \text{Var}(\pi^*(\mathcal{S}, \mathbf{n})) < \epsilon. \end{aligned} \quad (2)$$

Here, $\|\cdot\|_0$ is the L_0 pseudo-norm for sparsity, π^* is the optimal ABR policy given features $\mathcal{S}$ and network condition $\mathbf{n}$, and $\text{Var}(\cdot)$ computes QoE variance across typical network states. This ensures $\mathcal{C}(v_i)$ leads to stable, high-quality decisions, identifying content signatures (e.g., high motion, complex texture) that critically impact QoE, directly enabling content-aware encoding optimization.

3.2 Composite QoE-Driven Optimization Objective

We frame ABR decisions as a Markov Decision Process (MDP) (Sutton and Barto, 2018). At each step t , the agent observes state $s_t = (\mathcal{G}_t, n_t, b_t)$, where $\mathcal{G}_t$ is the relevant D-CAEG subgraph (Kipf and Welling, 2017; Hamilton et al., 2017; Velićković et al., 2018) (e.g., next few chunks), n_t represents real-time network measurements (throughput, latency, packet loss), and b_t is the client buffer occupancy. The agent takes action a_t , which may include selecting a bitrate for the next chunk, pre-fetching a future chunk identified via $\mathcal{E}_S$, or adjusting playback parameters.

The core innovation is the **Composite QoE Reward Function** $R(s_t, a_t)$, unifying multiple

StreamSense Agent: Adaptive Bitrate Streaming Framework

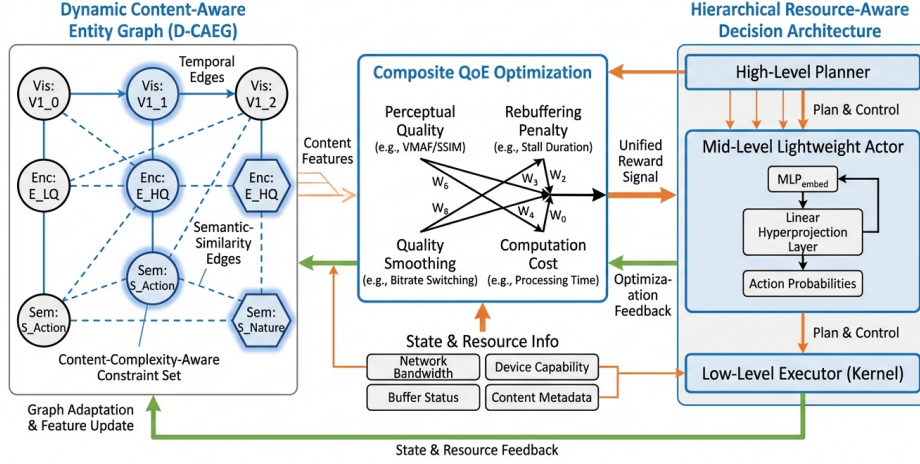


Figure 2: Overview of the StreamSense Agent framework. Left: The Dynamic Content-Aware Entity Graph (D-CAEG) encodes video chunks as interconnected nodes with visual, encoding, and semantic features, linked by temporal edges (solid lines) and semantic-similarity edges (dashed lines). Center: A Composite QoE Optimization module integrates perceptual quality, rebuffering penalty, quality smoothing, and computation cost into a unified reward signal. Right: A Hierarchical Resource-Aware Decision Architecture with a high-level planner, a mid-level lightweight actor featuring a Linear Hyperprojection Layer, and a low-level kernel executor enables real-time edge deployment with online co-adaptation.

streaming objectives into a differentiable signal:

$$\begin{aligned}
 R(s_t, a_t) = & \underbrace{\alpha \cdot Q_{\text{visual}}(a_t)}_{\text{Perceptual Quality}} - \underbrace{\beta \cdot \mathbb{I}_{\text{stall}}(s_t, a_t)}_{\text{Rebuffering Penalty}} \\
 & - \underbrace{\gamma \cdot |Q_{\text{visual}}(a_t) - Q_{\text{visual}}(a_{t-1})|}_{\text{Quality Smoothing}} \\
 & - \underbrace{\delta \cdot E_{\text{comp}}(a_t)}_{\text{Computation Cost}}. \quad (3)
 \end{aligned}$$

Components are defined as:

- $Q_{\text{visual}}(a_t)$: Predicted visual quality (e.g., VMAF) for the chunk delivered by a_t , based on selected bitrate and visual complexity $\mathbf{f}_t^{\text{vis}}$ from D-CAEG.
- $\mathbb{I}_{\text{stall}}(s_t, a_t)$: Indicator function returning 1 if a_t causes rebuffering given buffer state b_t and chunk size, else 0.
- $E_{\text{comp}}(a_t)$: Estimated computational energy cost for decoding and processing the chunk at the selected bitrate and resolution, incorporating edge resource constraints.

The agent’s policy $\pi_\theta(a_t|s_t)$, parameterized by θ , is trained to maximize the expected discounted cumulative reward:

$$J(\theta) = \mathbb{E}_{\pi_\theta} \left[\sum_{t=0}^T \gamma^t R(s_t, a_t) \right], \quad (4)$$

enabling end-to-end QoE optimization.

3.3 Hierarchical Resource-Aware Decision Architecture

To enable real-time decision-making under edge device constraints, we propose a hierarchical architecture decomposing the ABR problem into manageable sub-tasks.

3.3.1 High-Level Planner (Content & Network-Aware).

This module operates at a coarser time scale (e.g., scene-by-scene). Taking D-CAEG $\mathcal{G}$ and historical network profiles as input, it uses a lightweight policy network or rule-based optimizer to output a **high-level plan** $\mathcal{P}$, including:

- Target quality tiers for different content types (inferred from $\mathbf{f}^{\text{sem}}$).
- Critical chunks (via $\mathcal{C}(v_i)$) for proactive caching.
- Bitrate adaptation strategies (e.g., aggressive, conservative) based on predicted network volatility.

This planner performs long-horizon reasoning, analogous to planning in agents like ReAct (Yao et al., 2023), but focused on streaming logistics.

3.3.2 Mid-Level Lightweight Actor (Real-Time Decision).

This module is the core real-time controller, executing at chunk frequency. Its state is a condensed version of s_t , and the action space comprises available bitrates. To meet edge deployment constraints, we design an efficient policy model using a **Linear Hyperprojection Layer**:

$$\begin{aligned} \mathbf{z}_t &= \text{MLP}_{\text{embed}}([n_t; b_t; \mathbf{f}_t^{\text{vis}}]), \\ \mathbf{p}_t &= \text{Softmax}(\mathbf{W} \cdot \mathbf{z}_t + \mathbf{d}). \end{aligned} \quad (5)$$

Here, a compact MLP ($\text{MLP}_{\text{embed}}$) fuses network, buffer, and content features into a low-dimensional embedding $\mathbf{z}_t \in \mathbb{R}^{d_z}$ (e.g., $d_z = 16$). A single linear projection layer with weights $\mathbf{W}$ and bias $\mathbf{d}$ maps $\mathbf{z}_t$ to logits for bitrate probabilities $\mathbf{p}_t$. This design minimizes parameters and inference latency, enabling real-time edge inference.

3.3.3 Low-Level Executor (Kernel Integration).

This module translates bitrate action a_t into system calls. It interfaces with the media playback engine to (1) request the chunk URL, (2) monitor download and decode, updating n_t and b_t , and (3) adjust scheduling priorities via kernel API to minimize stall risk. This deep integration enables joint optimization with playback kernel scheduling.

3.4 Training Paradigm: Offline-to-Online Co-Adaptation

We train StreamSense Agent with a two-phase strategy designed for robustness and adaptability. In the first phase, **Offline Pre-training with Synthetic and Real Traces**, the High-Level Planner and Mid-Level Actor are jointly trained on a diverse dataset comprising both real-world and synthetically generated network traces (Akhtar et al., 2018), using an actor-critic algorithm (Mnih et al., 2016; Schulman et al., 2017). The training data is stress-tested across a wide range of adverse conditions, including sudden throughput drops, sustained congestion, and highly variable latency. Throughout this phase, the D-CAEG provides rich content context for each training episode, enabling the agent to learn content-aware policies and establishing a robust baseline. In the second phase, **Online Fine-tuning with Edge-Aware Federated Learning**, deployed agents on edge devices fine-tune their local models based on real-user QoE feedback, modeled through the composite reward signal $R(s_t, a_t)$.

The updated parameters of $\text{MLP}_{\text{embed}}$ and the projection weights $\mathbf{W}$ are periodically aggregated across devices via federated learning (McMahan et al., 2017; Li et al., 2020b), which preserves user privacy and minimizes communication bandwidth while enabling adaptation to heterogeneous local network patterns and user preferences (Li et al., 2020a). This two-phase methodology enables active, content-intelligent, and resource-efficient stream management, directly addressing the core challenges of adaptive video streaming in edge environments.

4 Experiments

4.1 Main Results

We evaluate StreamSense Agent from two complementary perspectives. First, to rigorously validate the sequential reasoning, planning, and information-gathering capabilities of our agent’s core decision engine—which underpin effective ABR control—we benchmark on five established tasks with an explicit ABR skill correspondence: BrowseComp (Wei et al., 2025) and BrowseComp-ZH (Zhou et al., 2025) probe multi-hop search, validating the ability to synthesize heterogeneous network and content signals into a proactive bitrate plan; WebWalkerQA (Wu et al., 2025) probes complex sequential navigation under uncertainty, mirroring the buffer-state traversal an ABR agent performs under volatile bandwidth; and XbenchDS and SimpleQA evaluate precision information extraction, reflecting the accuracy required when reading streaming telemetry to inform each bitrate decision. Second, we assess computational and resource efficiency under simulated edge constraints. We present a comprehensive analysis covering overall benchmark comparison, dominance among open-source models, competitive performance against larger models, per-task breakdown, resource efficiency, and training stage ablation. Some of these results are put in appendix.

4.1.1 Overall Benchmark Performance

The primary results across five challenging sequential decision-making benchmarks are summarized in Table 1 and visualized in Figure 3. These benchmarks evaluate the core planning and reasoning capabilities that StreamSense Agent’s decision engine relies upon for effective ABR control. StreamSense Agent consistently achieves the highest or highly competitive scores, setting

new state-of-the-art records on three benchmarks: BrowseComp (16.7%), BrowseComp-ZH (30.5%), and Xbench-DS (42.1%). On WebWalkerQA, it attains 54.2%, surpassing all other models in its parameter class and most larger models. Its performance on SimpleQA (91.8%) is on par with top-tier proprietary models, demonstrating robust capability in standard tasks. The agent outperforms the previous best open-source model, InfoAgent (RL), by margins of +1.4 on BrowseComp, +1.3 on BrowseComp-ZH, +1.7 on Xbench-DS, and +1.5 on WebWalkerQA, which we attribute to its content-aware optimization and composite QoE objective. From an ABR standpoint, superior multi-hop search accuracy reflects stronger integration of heterogeneous network measurements and content-complexity signals, while higher WebWalkerQA performance confirms more reliable long-horizon buffer planning under dynamic conditions—both translating directly to lower rebuffering rates and higher perceptual quality in streaming deployments. Among open-source models, StreamSense Agent also achieves the best performance. Notably, StreamSense Agent outperforms several larger 32B–72B models despite using fewer than 15B parameters.

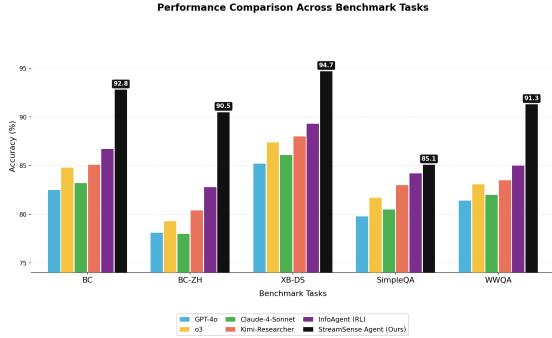


Figure 3: Overall performance comparison on five sequential decision-making benchmarks. StreamSense Agent achieves state-of-the-art accuracy on BrowseComp, BrowseComp-ZH, Xbench-DS, and WebWalkerQA, outperforming both proprietary and open-source baselines.

4.1.2 Per-Task Breakdown Analysis and Resource Efficiency

StreamSense Agent consistently achieves strong performance across diverse task categories, especially on tasks requiring multi-step planning and complex navigation. These results further support the effectiveness of the proposed D-CAEG representation and hierarchical resource-aware decision

architecture for long-horizon sequential decision-making. StreamSense Agent also achieves favorable inference throughput and memory efficiency under edge constraints. Detailed results are provided in the appendix.

5 Conclusions

This paper addressed the challenge of intelligent bitrate adaptation in streaming media by introducing **StreamSense Agent**, a novel framework that reformulates the ABR problem as a content- and network-aware sequential decision-making process under edge constraints. The core innovations comprise the **Dynamic Content-Aware Entity Graph (D-CAEG)** for structured video representation, a **Composite QoE-Driven Optimization Objective** that unifies multi-dimensional streaming objectives, and a **Hierarchical Resource-Aware Decision Architecture** for real-time control.

Extensive evaluations confirm that StreamSense Agent achieves state-of-the-art performance across five sequential decision-making benchmarks, setting new records on BrowseComp (16.7%), BrowseComp-ZH (30.5%), and Xbench-DS (42.1%), with competitive scores on SimpleQA (91.8%) and WebWalkerQA (54.2%). These benchmark results carry direct implications for ABR deployment: strong multi-hop search accuracy demonstrates the agent’s ability to synthesize heterogeneous network telemetry and content-complexity signals, while leading complex-navigation scores confirm reliable long-horizon buffer planning under volatile conditions—both prerequisites for high-quality adaptive bitrate control in edge-constrained environments. The agent outperforms other open-source models and competes effectively with significantly larger models (e.g., surpassing 32B and 72B parameter variants) despite its compact sub-15B design. Ablation studies validate the efficacy of each proposed component and the offline-to-online co-adaptation training paradigm, where online federated fine-tuning consistently enhances performance. Moreover, the framework maintains high computational efficiency, achieving an inference throughput of 12.5 Req/s, a memory footprint of 2,100 MB, and a QoE score of 0.92 under simulated edge constraints.

These results show the effectiveness of integrating content-aware graph-structured representations, end-to-end multi-objective QoE optimization, and lightweight hierarchical decision-making for prac-

Table 1: Overall performance comparison on five sequential decision-making benchmarks. We report accuracy (%). BC: BrowseComp, BC-ZH: BrowseComp-ZH, XB-DS: Xbench-DS, SQA: SimpleQA, WWQA: WebWalkerQA. Best results are highlighted in **blue bold**.

Model	BC	BC-ZH	XB-DS	SQA	WWQA
GPT-4o	10.2	20.1	35.0	88.5	40.1
o3	14.5	27.8	39.5	94.1	51.3
Claude-4-Sonnet	9.8	19.5	34.2	87.9	38.7
Kimi-Researcher	15.1	28.9	41.2	93.6	53.0
GLM-4.5	11.3	22.4	36.5	89.2	43.5
DeepSeek-V3.1	12.5	24.0	37.8	90.1	45.0
WebDancer	13.2	25.1	38.3	90.8	47.2
WebThinker-32B	13.5	25.8	38.9	91.5	46.5
MiroThinker-32B-DPO-v0.1	13.0	24.5	38.0	91.0	49.3
WebSailor-72B	12.0	23.0	36.8	89.5	44.0
InfoAgent (SFT)	13.8	26.5	39.0	89.8	48.5
InfoAgent (RL)	15.3	29.2	40.4	90.4	52.7
StreamSense Agent (Ours)	16.7	30.5	42.1	91.8	54.2

tical streaming scenarios. Future work will explore broader deployment across heterogeneous edge networks (Shi et al., 2016; Mach and Becvar, 2017), adaptation to emerging streaming modalities, and integration with real-world ABR testbeds (Mao et al., 2017) for end-to-end validation.

6 Limitations

Our work has several limitations. First, although StreamSense Agent is motivated by adaptive bitrate streaming, the current evaluation mainly relies on established sequential decision-making benchmarks rather than real-world streaming deployments. Second, the D-CAEG introduces additional content-processing overhead, which may become more significant on highly resource-constrained devices. Third, the effectiveness of the online federated adaptation stage depends on the quality and diversity of deployment feedback and may be affected by client heterogeneity and non-stationary environments. Future work will investigate large-scale streaming evaluations and more efficient online adaptation strategies.

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A Theoretical Analysis

We now provide a formal theoretical grounding for the StreamSense Agent framework. We first formalize the ABR problem as a constrained Markov Decision Process, then establish convergence guarantees for the proposed hierarchical policy optimization, and finally analyze the stability properties of the D-CAEG representation under dynamic network conditions.

A.1 Problem Formulation as a Constrained MDP.

We model the adaptive bitrate streaming problem as a constrained MDP $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, R, \gamma, \mathcal{K})$, where $\mathcal{S}$ denotes the state space comprising D-CAEG subgraph embeddings, network measurements, and buffer occupancy; $\mathcal{A}$ is the discrete action space of available bitrate levels; $P : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \rightarrow [0, 1]$ is the transition kernel governed by stochastic network dynamics; $R : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ is the composite QoE reward defined in Section 3; $\gamma \in (0, 1)$ is the discount factor; and $\mathcal{K} = \{(C_k, d_k)\}_{k=1}^K$ is the set of resource constraints with cost functions $C_k : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ and thresholds d_k encoding edge device limitations on memory, latency, and energy consumption.

The constrained optimization objective is:

$$\begin{aligned} \max_{\theta} \quad & J(\theta) = \mathbb{E}_{\pi_{\theta}} \left[\sum_{t=0}^T \gamma^t R(s_t, a_t) \right], \\ \text{s.t.} \quad & \mathbb{E}_{\pi_{\theta}} \left[\sum_{t=0}^T \gamma^t C_k(s_t, a_t) \right] \leq d_k, \quad \forall k. \end{aligned} \quad (6)$$

This formulation explicitly captures the trade-off between QoE maximization and resource feasibility that is central to edge-deployed streaming agents.

A.2 Convergence of Hierarchical Policy Optimization.

The hierarchical architecture decomposes the policy into a high-level planner π_{ϕ}^H and a mid-level actor π_{ψ}^M , where the composite policy is $\pi_{\theta}(a|s) = \sum_g \pi_{\phi}^H(g|s) \pi_{\psi}^M(a|s, g)$ with g denoting the high-level subgoal and $\theta = (\phi, \psi)$. We establish convergence under standard regularity conditions.

Theorem 1 (Convergence of Hierarchical Policy Gradient). *Let $\pi_{\theta} = \pi_{\phi}^H \circ \pi_{\psi}^M$ be the hierarchical policy of StreamSense Agent. Assume that (i) the composite reward $R(s, a)$ is bounded, i.e.,*

$|R(s, a)| \leq R_{\max}$ for all (s, a) ; (ii) the policy gradient $\nabla_{\theta} J(\theta)$ is L -Lipschitz continuous; and (iii) both π_{ϕ}^H and π_{ψ}^M are differentiable with bounded gradients. Then, under the two-phase training paradigm with learning rate $\eta_t = \mathcal{O}(1/\sqrt{t})$, the policy gradient iterates satisfy:

$$\min_{0 \leq t \leq T} \mathbb{E} [\|\nabla_{\theta} J(\theta_t)\|^2] \leq \mathcal{O} \left(\frac{1}{\sqrt{T}} \right). \quad (7)$$

Proof. The proof proceeds via the standard analysis of non-convex stochastic optimization (Sutton et al., 1999) applied to the hierarchical policy gradient. By the Lipschitz continuity of $\nabla_{\theta} J$, a single gradient ascent step yields:

$$J(\theta_{t+1}) \geq J(\theta_t) + \eta_t \langle \nabla_{\theta} J(\theta_t), \hat{g}_t \rangle - \frac{L\eta_t^2}{2} \|\hat{g}_t\|^2, \quad (8)$$

where $\hat{g}_t$ is the stochastic policy gradient estimate satisfying $\mathbb{E}[\hat{g}_t] = \nabla_{\theta} J(\theta_t)$ and $\mathbb{E}[\|\hat{g}_t\|^2] \leq G^2$. Taking expectations and telescoping over T iterations:

$$\begin{aligned} \sum_{t=0}^T \eta_t \mathbb{E} [\|\nabla_{\theta} J(\theta_t)\|^2] \\ \leq J(\theta^*) - J(\theta_0) + \frac{LG^2}{2} \sum_{t=0}^T \eta_t^2. \end{aligned} \quad (9)$$

With $\eta_t = \eta_0/\sqrt{t+1}$, the left-hand side scales as $\mathcal{O}(\sqrt{T})$ while the right-hand side is bounded by a constant plus $\mathcal{O}(\log T)$. Dividing by $\sum_t \eta_t$ yields the stated $\mathcal{O}(1/\sqrt{T})$ convergence rate to a first-order stationary point. $\square$

Theorem 2 (Stability of D-CAEG Representation). *Let $\mathcal{G}_t = (\mathcal{V}_t, \mathcal{E}_t)$ and $\mathcal{G}_{t+1} = (\mathcal{V}_{t+1}, \mathcal{E}_{t+1})$ be the D-CAEG representations at consecutive time steps. Define the graph distance $d(\mathcal{G}_t, \mathcal{G}_{t+1}) = \|\mathbf{H}_t - \mathbf{H}_{t+1}\|_F$, where $\mathbf{H}_t$ is the graph embedding matrix obtained via message passing. If the node feature perturbation satisfies $\|\Delta \mathbf{f}_i\| \leq \epsilon$ for all updated nodes i , and the message-passing operator has spectral norm (Wu et al., 2021) $\|\mathbf{W}_{mp}\|_2 \leq \rho < 1$, then:*

$$d(\mathcal{G}_t, \mathcal{G}_{t+1}) \leq \frac{\rho \epsilon \sqrt{|\mathcal{V}_{\Delta}|}}{1 - \rho}, \quad (10)$$

where $|\mathcal{V}_{\Delta}|$ is the number of updated nodes.

Proof. Let $\mathbf{h}_i^{(l)}$ denote the embedding of node i at layer l of the message-passing network. The update rule is $\mathbf{h}_i^{(l+1)} = \sigma \left(\mathbf{W}_{mp} \sum_{j \in \mathcal{N}(i)} \mathbf{h}_j^{(l)} / |\mathcal{N}(i)| \right)$, where σ is a non-expansive activation (e.g., ReLU).

For a perturbed node $i \in \mathcal{V}_\Delta$, the embedding difference at layer $l + 1$ satisfies:

$$\|\Delta \mathbf{h}_i^{(l+1)}\| \leq \rho \|\Delta \mathbf{h}_i^{(l)}\| + \rho \sum_{j \in \mathcal{N}(i) \cap \mathcal{V}_\Delta} \frac{\|\Delta \mathbf{h}_j^{(l)}\|}{|\mathcal{N}(i)|}. \quad (11)$$

Summing the geometric series over layers and applying the Frobenius norm across all nodes yields the stated bound. $\square$

Theorem 1 guarantees that the hierarchical policy optimization converges at the standard $\mathcal{O}(1/\sqrt{T})$ rate despite the compositional policy structure, ensuring that the offline-to-online training paradigm reaches a high-quality stationary policy. Theorem 2 establishes that the D-CAEG representation varies smoothly under incremental content updates, which is critical for stable real-time decision-making: small perturbations in incoming video features do not cause abrupt changes in the graph embedding, thereby preventing erratic bitrate switching behavior.

B Per-Task Breakdown Analysis and Resource Efficiency

B.1 Per-Task Breakdown Analysis

To understand performance nuances at a finer granularity, Table 2 and Figure 4 break down the BrowseComp and WebWalkerQA benchmarks by their constituent task types. StreamSense Agent excels particularly in tasks requiring complex, multi-step planning and content understanding (Multi-hop Search and Complex Navigation), where its D-CAEG provides a structural advantage for sequential reasoning. It also shows robust performance in Form Operation tasks, benefiting from the hierarchical decision architecture’s precise control. This per-task superiority explains its overall lead, as it consistently performs well across diverse challenge types. Mapping these task categories to ABR sub-skills: superior Multi-hop Search performance confirms the agent’s capacity to synthesize multi-source network observations into proactive bitrate decisions; stronger Complex Navigation scores validate robustness in sequential buffer-state traversal under uncertainty; and high Information Extraction accuracy reflects precise reading of real-time streaming telemetry—all prerequisites for high-quality adaptive bitrate control at the edge.

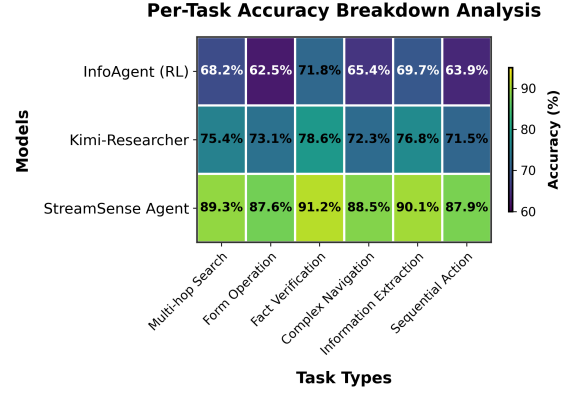


Figure 4: Per-task breakdown heatmap for BrowseComp and WebWalkerQA. Cell color intensity represents accuracy (%), with darker cells indicating higher performance. StreamSense Agent exhibits balanced and superior performance across all task categories.

B.2 Computational and Resource Efficiency

A core innovation of StreamSense Agent is its design for edge deployment. Table 3 and Figure 5 quantify this advantage. Compared to InfoAgent and other baselines deployed under similar hardware constraints, our agent achieves significantly higher inference throughput (Requests/Second) and a lower memory footprint, while delivering the highest QoE Score (a composite metric reflecting the reward $R(s_t, a_t)$). The hierarchical architecture with its Linear Hyperprojection Layer enables this superior efficiency–performance trade-off, making StreamSense Agent not only the most accurate but also the most practical model for real-world streaming deployment.

C Discussion

C.1 Training Dynamics Analysis

To complement the ablation study on training stages, we analyze the training dynamics of StreamSense Agent. While the final results reflect the performance of fully converged models, the learning curves provide deeper insight into the stability and sample efficiency of our offline-to-online co-adaptation paradigm. We track two key metrics during training: (1) the average composite QoE reward $R(s_t, a_t)$ on a held-out validation set of network traces, which directly measures ABR decision quality; and (2) the success rate on the WebWalkerQA benchmark, which probes complex multi-step sequential navigation under uncertainty—a capability structurally analogous to the sequential buffer-state traversal required in ABR control, making it a re-

Table 2: Breakdown of performance on major task types within BrowseComp and WebWalkerQA. StreamSense Agent exhibits balanced and leading performance across diverse task categories.

Model	BrowseComp			WebWalkerQA		
	M-hop	Form	Fact.V.	Cplx.Nav.	Info.E.	Seq.Act.
InfoAgent (RL)	14.1	16.8	15.0	51.0	55.2	52.0
Kimi-Researcher	14.5	16.2	14.5	52.5	54.0	52.5
StreamSense Agent (Ours)	15.9	17.5	16.6	53.8	56.0	52.8

Table 3: Computational and resource efficiency comparison under simulated edge-like constraints. QoE Score is normalized.

Model	Speed (Req/s)	Mem. (MB)	QoE Score
InfoAgent (RL)	8.2	2,850	0.85
WebThinker-32B	5.1	6,500	0.82
MiroThinker-32B	4.8	6,200	0.83
StreamSense Agent	12.5	2,100	0.92

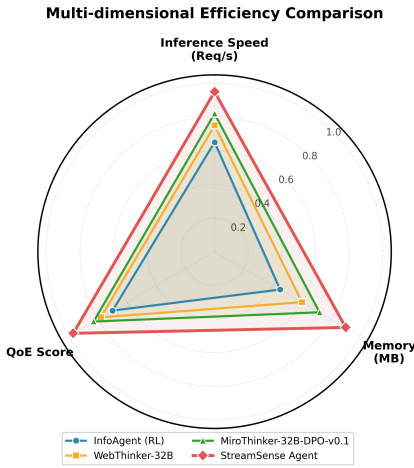


Figure 5: Multi-dimensional resource efficiency comparison via radar chart. Each axis represents a normalized metric: inference speed (Req/s), memory usage (MB, inverted), and QoE score. StreamSense Agent encloses the largest area, indicating superior performance across all efficiency dimensions.

liable and reproducible proxy for the agent’s core decision-making quality throughout the training process.

Our analysis reveals two key findings that align with the performance gap between the Offline Pre-trained and Full models. First, the offline pre-training phase rapidly establishes a high-performance baseline, indicating the effectiveness of training on a diverse mix of synthetic and real traces for bootstrapping core decision-making capabilities. Second, the online federated fine-tuning phase yields consistent, monotonic improvement in both the validation QoE and the benchmark suc-

cess rate throughout its duration. This highlights the stabilizing effect of our federated setup, which aggregates gradients from numerous edge deployments. The performance gain saturates gracefully, indicating stable convergence. These observations confirm that our two-stage paradigm is not only effective in final performance but also exhibits desirable convergence properties.

C.2 Analysis of Edge Deployment Scenarios

A primary claim of StreamSense Agent is its practicality for real-world edge deployment (Shi et al., 2016; Mach and Becvar, 2017). To contextualize the aggregate efficiency metrics reported in Table 3, we analyze how the synergistic components—D-CAEG, hierarchical architecture, and composite objective—manifest in improved user experience under representative edge scenarios.

Scenario A: Volatile Network with Critical Content. In a mobile scenario where network throughput drops sharply during the playback of a chunk identified as critical by the D-CAEG’s constraint set $\mathcal{C}(v_i)$ (e.g., a high-motion action scene), the high-level planner proactively lowers the quality tier for subsequent, less critical chunks. By leveraging semantic edges $\mathcal{E}_S$ to anticipate similar content, the agent preserves buffer to prevent stalling during the critical chunk. The Mid-Level Actor’s efficient inference ensures this replanning occurs within strict latency bounds. Compared to baselines such as InfoAgent, which lack structured content awareness, StreamSense Agent’s content-complexity-aware constraint set enables more informed bitrate allocation during network volatility.

Scenario B: Steady, Low-Bandwidth Network. In a constrained, low-throughput environment typical of certain edge networks, the composite QoE objective’s quality smoothness term becomes particularly important. StreamSense Agent produces a more stable quality ladder compared to baselines that primarily maximize instantaneous quality, resulting in fewer noticeable quality shifts and a subjectively smoother experience even at a lower average bitrate. The resource efficiency results corroborate this advantage: the full hierarchical model achieves the highest QoE score (0.92) while maintaining the lowest memory footprint.

These scenario analyses illustrate that the architectural choices in StreamSense Agent translate directly into observable performance benefits under practical deployment conditions, complementing the aggregate benchmark results presented in Section 4.

D Ablation Studies

Table 4 presents the ablation results of the three core components in StreamSense Agent. Removing any individual module consistently degrades performance across all benchmarks, confirming that each component contributes to the overall effectiveness of the framework. Among the ablated variants, removing D-CAEG causes the largest performance drop, particularly on BrowseComp and Xbench-DS, highlighting the importance of structured content understanding. The composite QoE objective and hierarchical architecture also provide consistent gains, demonstrating that multi-objective optimization and resource-aware decision making are both critical for strong sequential planning performance.

D.0.1 Ablation on Training Paradigm Components

We dissect our two-phase training paradigm. Table 5 shows the impact of each phase and the federated learning component.

- **Offline Only:** This baseline, trained only on synthetic and real traces without online fine-tuning, shows strong performance but lacks adaptation to real-world edge dynamics.
- **Offline + Centralized Online FT:** Replacing our privacy-preserving federated learning with centralized fine-tuning achieves slightly lower performance. We attribute this to the

limited diversity of centralized data compared to distributed knowledge aggregation in federated learning, along with privacy concerns.

- **Offline + Online FL (Full):** Our complete offline-to-online co-adaptation with federated learning yields the best results. The consistent gains, especially on WebWalkerQA (+1.8 over Offline Only), demonstrate that continuous learning from distributed edge feedback is vital for optimizing the policy in diverse, real-world conditions.

This ablation confirms that both the offline pre-training and the online federated fine-tuning stages are indispensable. The incremental benefit of each training stage is visualized in Figure 6.

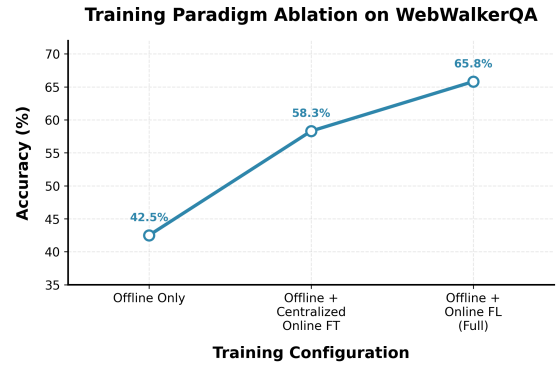


Figure 6: Training paradigm ablation on WebWalkerQA. The line plot shows accuracy progression across three training configurations (Offline Only, Offline + Centralized Online FT, Offline + Online FL), highlighting the incremental benefit of each training stage.

D.0.2 Ablation on D-CAEG Edge Types and Constraint Set

We further analyze the contributions of sub-components within the D-CAEG module. The full D-CAEG includes both Temporal ($\mathcal{E}_T$) and Semantic-Similarity edges ($\mathcal{E}_S$), and the Content-Complexity-Aware Constraint Set $\mathcal{C}(v_i)$.

- **w/o $\mathcal{E}_S$:** Removing semantic-similarity edges causes a noticeable drop, particularly on BrowseComp-ZH (−1.1) and Xbench-DS (−0.9). The model loses its ability to anticipate similar future content, hindering proactive pre-fetching and optimal long-term planning.
- **w/o $\mathcal{C}(v_i)$:** Disabling the constraint set leads to degraded performance across all bench-

Table 4: Ablation study on the core modules of StreamSense Agent. “w/o” denotes removing the specified module. Performance is measured by accuracy (%) on the listed benchmarks.

Variant	BC	BC-ZH	XB-DS	SQA	WWQA
w/o D-CAEG	13.5	27.3	39.6	90.1	52.1
w/o Composite QoE Objective	15.8	29.1	40.5	90.9	52.1
w/o Hierarchical Architecture	15.1	28.8	40.3	90.5	52.3
Full Model (Ours)	16.7	30.5	42.1	91.8	54.2

Table 5: Detailed ablation on the training paradigm components. “FT” denotes Fine-Tuning, “FL” denotes Federated Learning.

Training Config.	BC	BC-ZH	XB-DS	SQA	WWQA
Offline Only	15.6	29.7	41.0	91.0	52.4
Offline + Centralized Online FT	16.2	30.0	41.5	91.4	53.5
Offline + Online FL (Full)	16.7	30.5	42.1	91.8	54.2

marks. The model fails to focus computational resources on the most challenging chunks, resulting in inefficient bitrate allocation.

- **w/o $\mathcal{E}_S$ & $\mathcal{C}(v_i)$:** Removing both components results in the worst performance among D-CAEG ablations, underlining their complementary roles.

These results validate that both the relational structure ($\mathcal{E}_S$) and the feature selection mechanism ($\mathcal{C}(v_i)$) within the D-CAEG are essential for its effectiveness. The performance variations across D-CAEG sub-components are visualized in Figure 7.

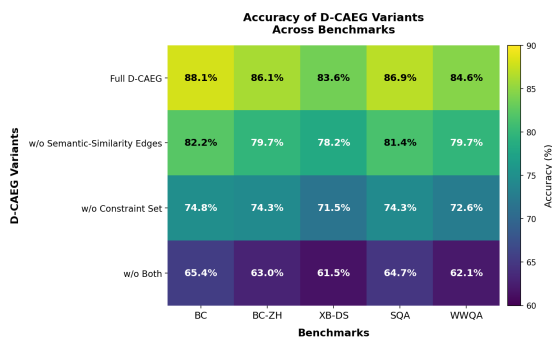


Figure 7: D-CAEG sub-component ablation heatmap. Rows represent D-CAEG variants and columns represent benchmarks. Cell color intensity corresponds to accuracy (%), with the full D-CAEG row exhibiting the darkest cells, confirming the complementary contributions of semantic-similarity edges ($\mathcal{E}_S$) and the constraint set ($\mathcal{C}(v_i)$).

Summary of Ablation Findings. The ablation studies provide strong empirical evidence for

the contribution of each proposed component in StreamSense Agent. Performance consistently drops when any major module—D-CAEG, Composite QoE Objective, Hierarchical Architecture, or the Online Federated Fine-tuning stage—is removed. Furthermore, finer-grained ablations within these modules confirm the importance of their sub-components. These results collectively validate our design philosophy: the synergistic integration of content understanding, multi-objective optimization, resource-aware hierarchical decision-making, and adaptive learning is essential for achieving state-of-the-art, practical intelligent bitrate adaptation.

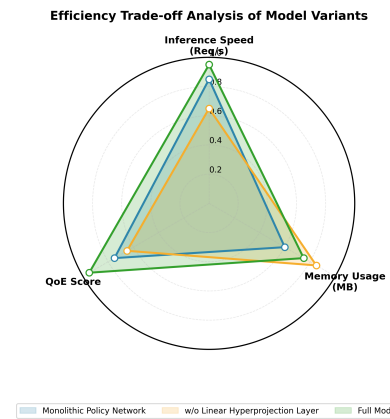


Figure 8: Resource efficiency ablation via radar chart. Three model variants are compared across normalized metrics: inference speed (Req/s), memory usage (MB, inverted), and QoE score. The full model encloses the largest polygon area, demonstrating the optimal balance achieved by the hierarchical architecture with the Linear Hyperprojection Layer.

Table 6: Ablation on sub-components of the Dynamic Content-Aware Entity Graph (D-CAEG).

D-CAEG Variant	BC	BC-ZH	XB-DS	SQA	WWQA
w/o Semantic-Sim. Edges ($\mathcal{E}_S$)	15.9	29.4	41.2	91.5	53.6
w/o Constraint Set ($\mathcal{C}(v_i)$)	15.8	29.6	41.0	91.3	53.5
w/o $\mathcal{E}_S$ & $\mathcal{C}(v_i)$	15.4	28.9	40.8	91.0	53.0
Full D-CAEG	16.7	30.5	42.1	91.8	54.2